

Sai Anoop Avunuri

847-630-3889 || saianoop9@gmail.com || [linkedin.com/in/sai-anoop](https://www.linkedin.com/in/sai-anoop) || [GitHub](#) || [Portfolio](#)

EDUCATION

Georgia Institute of Technology

Aug 2021 – May 2025

Master's Degree: Computer Science, Machine Learning specialization || **GPA:** 4.00/4.00

Bachelor's Degree: Computer Science || **GPA:** 4.00/4.00

Awards/Honors: 1st place @ Hacklytics hackathon (1000+ participants), 1st place @ HackGSU hackathon, National Merit Scholar

TECHNICAL SKILLS

Programming Languages: Go, Python, Rust, SQL, Java, C, C++, JavaScript, Bash, Flutter, Assembly

Infrastructure & Cloud: AWS, GCP, Kubernetes, Docker, Pulumi, FaaS, Linux, Git, Bazel, Caddy, Datadog

Databases & Systems: Redis, Postgres, MongoDB, Kafka, MySQL, Hive, S3

Frameworks: PyTorch, Keras, Langchain, Pandas, Numpy, Django, Node.js, REST, Figma, Bootstrap, Ethereum.py

Core Competencies: Distributed Systems, Data Structures & Algorithms, Databases, Operating Systems, Networking, Object-Oriented Design, Machine Learning, Reinforcement Learning, Computer Vision, Artificial Intelligence, Probability & Statistics

EXPERIENCE

TikTok || Backend Software Engineer || Creator Strategy - Subscription

June 2025 - Present

- Owned the **SDLC for Account Governance System** safeguarding a **30M+ creator** ecosystem; recovered **\$440K** & revoked access to **1700+** fraudulent creators; partnered with governance and payment teams to create **scalable, async** workflows
- Spearheaded an **event-driven pipeline** using an **EventBus** and **Hive DW tables**, processing reviews using **RPC** callbacks
- Built a **distributed Finite State Machine (FSM)** for user moderation; leveraged a **DAG-based transactional framework (ByteState)** to guarantee **eventual consistency** across microservice databases; enforced **idempotency** in the pipeline steps
- Architected a **high-throughput database migration** to a new schema handling **>50K DB QPS** and **>1M cache QPS**
- Boosted development velocity and accelerated feature delivery by leveraging **Spec-Driven Development (SDD)** to ground AI coding models, while utilizing **Git worktrees** to parallelize implementation of 3 sub-features
- Led a **zero-downtime** monorepo split using **Bazel** to isolate core services, authoring rigorous SOPs for rollouts and rollbacks
- Accelerated business growth by developing the sub-only photos feature, resulting in a **~7% increase in GMV**
- Optimized **system observability** by mapping **CUJs** to **metric dashboards** and creating **targeted SLO violation alarms**
- Managed on-call rotations, diagnosing and solving issues in infra (**RPC** timeouts, **SLA** drops, pod failures) and logic bugs

Pinecone || SWE Intern || Experience Team

May 2024 – Aug 2024

- Designed & developed a **clustering** service enabling users to **partition high-dimensional data** and extract insights
- Implemented **REST** endpoints and DB schemas using **Postgres** and **AWS S3**, implemented a core operational scheduler to handle **pagination** and status tracking, added **Datadog** monitoring, and implemented **k-means clustering** on the data
- Orchestrated a **Kubernetes** cluster for container deployment, utilizing **Pulumi** to automate scalable cloud service deploys
- Executed comprehensive **stress testing** on **100K+ size datasets** to identify and resolve memory and runtime bottlenecks

RxSense || Data Science Intern || Analytics Team

Jan 2023 – Aug 2023

- Profiled **execution bottlenecks** using **cProfile**, refactoring the analytics engine to eliminate high-overhead NumPy object read operations via optimized dictionary lookups, achieving a **30x runtime performance increase**
- Developed an **ML-based analytics dashboard** to analyze drug transaction types and quantities, driving strategic insights
- Automated several manual tasks using **Python** scripts such as Jira ticket creation and Slack messaging, saving 30 mins daily

PROJECTS

Sharded Key-Value Store || Distributed Systems

May 2025

- Developed a **distributed key-value service** replicated via the **Raft consensus algorithm** to enforce **strong consistency**
- Designed a **dynamic sharding architecture** and configuration manager to balance loads and migrate shards across nodes
- Validated the service for **linearizability** and **low latency**, against unreliable network, server crashes, and slow replicas

ML for disease pathway prediction || 1st Place Winner at Hacklytics 2025

Feb 2025

- Conceptualized an **end-to-end predictive tool** utilizing **ESM2** embeddings and **AlphaFold** structures to map human protein-protein interactions (PPI) and visualize disease pathways in 3D
- Constructed the **data preprocessing pipeline**, curating a heavily **one-hot encoded** training dataset and mitigating extreme class imbalance via a custom **weighted Binary Cross-Entropy (BCE)** loss function

Adversarial ML for ResNet || ML Security

Feb 2025

- Crafted a **Projected Gradient Descent** adversarial attack to achieve **99% misclassification accuracy** against **ResNet**
- Implemented **adversarial training loops** in **PyTorch**, elevating classification accuracy to 93% under active threat vector

ProteinCLIP++ || Computational Biology & Deep Learning

March 2025 – Apr 2025

- Designed and trained a **multi-modal tri-contrastive learning model** mapping structural, sequence, and functional data
- Optimized **training pipeline efficiency**, outperforming the baseline model while utilizing only **2% of reference dataset**